Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/20/2025
PI Name:	Raithel, Georg Ph.D.		
Project Title:	High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-1040	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	08/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/31/2021 per U. Israelsson/JPL (Ed., 1/5/21) NOTE: End date changed to 8/31/2020 per U. Israelsson/JPL (Ed., 8/6/19)		
Key Personnel Changes/Previous PI:	2025 Final report: There were no changes in PI and CoI through the lifetime of the grant.		
COI Name (Institution):	Malinovsky, Vladimir Ph.D. (Stevens Institute of Technology)		
Grant/Contract No.:	JPL 1501691		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Objective: The system of fundamental constants (the speed of light, Planck's constant, the fine structure constant, the Rydberg constant, etc.) is important because it directly reflects our current understanding of the foundations of nature. The objective of the project is to measure one of these constants, the Rydberg constant, at an uncertainty level in the range of a few parts in 10^12 (the current uncertainty stands at 5.9 parts in 10^12). It is planned to use circular Rydberg atoms, which are very large, highly excited atoms with doughnut-shaped electron distributions that do not penetrate into the atomic nucleus. As a result, these atoms are not susceptible to perturbations caused by electron wavefunction overlap with the nuclear charge density and quantum-electrodynamics (QED) corrections. The measurement therefore yields a result for the Rydberg constant that is not contingent to the accuracy/precision of input from other areas of physics. It is this property that makes a measurement of the Rydberg constant with circular Rydberg atoms particularly useful, as it will present a valuable complement/check for results based on established methods. Due to the absence of critical QED corrections and nuclear penetration, a measurement of the Rydberg constant with circular Rydberg atoms would help resolve a recently reported inconsistency, where it was concluded that "... either the Rydberg constant has to be shifted by -110kHz/c (4.9 standard deviations), or the calculations of the QED [quantum-electro-dynamic] effects in atomic hydrogen or muonic hydrogen atoms are insufficient." [R. Pohl et al., Nature 466, 213 (2010)]. Method: Rb atoms are cooled to below 1 microK and laser-excited into a low-angular-momentum Rydberg level with principal quantum number n on the order of 40. Using an adiabatic atomic-state transformation method, the Rydberg atoms are then transferred into the circular state |n,n1=0,n2=0,m=n-1> (m, n1, and n2 are the magnetic and parabolic quantum numbers). An optical lattice (wavelength 1064 nm or 532 nm on the ground, 850 nm at CAL) is adiabatically ramped on. The Rydberg atoms are trapped in the lattice via the ponderomotive interaction of the Rydberg electron with the lattice light. During the hold time of the atom cloud in the lattice, the electric-quadrupole transition into the state |n+2,n1=1,n2=1,m=n-1> is driven by amplitude-modulating the optical lattice at the transition frequency. The transition is probed using state-selective electric-field ionization. The measured transition frequency allows one to extract the Rydberg constant. Key advantages of the method are that the transition is free of first-order Stark and Zeeman shifts, that QED, nuclear-penetration, hyperfine, and other shifts are small, and that the optical lattice causes only moderate trap-induced transition shifts. Theoretical work includes the modeling of the circular-state preparation using multi-level adiabatic passage and of the circular-state transition dynamics in a shallow, microwave-modulated optical lattice, in which center-of-mass wave-packet expansion and tunneling must be considered. Relevance to NASA and relevance of microgravity: High-precision measurement of fundamental constants represents an important part of contemporary atomic-physics research. The proposed high-precision measurement of the Rydberg constant adds a new component to the NASA Fundamental Physics program, and to the Cold Atom Laboratory (CAL) research portfolio. To achieve optimal spectroscopic performance it is critical to reduce systematic lattice-induced transition shifts and related shifts. Since the most critical shifts diminish with decreasing optical-lattice depth, the lattices must be chosen as shallow as possible. Microgravity conditions are important because they enable the use of shallow lattices without the atoms falling out. The microgravity environment provided by the CAL/ its successor instrument, including its 850 nm optical-lattice laser, offers an ideal platform for the project to reach its ultimate uncertainty goal. An International Space Station (ISS)-based setup could also provide a sufficient period of measurement time to acquire good statistics in the high-precision spectra. More generally, the research could bring Rydberg atoms into space. This has far-reaching additional potential for the development of novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of the physics of very dilute, but strongly interacting Bose Einstein Condensates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	High-precision spectroscopy and metrology impacts fields such as time measurement and navigation. Benefits to the general public include atomic clocks and the GPS system. The project "High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity" involves the development of a fundamentally new type of spectroscopy, which has been successfully demonstrated during the first year of the grant. The future development of laser systems to hone this innovative technique may lead to benefits in other disciplines, such as chemistry, biology, and engineering. The research involves graduate students, undergraduate students, and post-docs, who are being prepared for future jobs in academia, government, and industry. Having a capable scientific workforce is essential for progress and competitiveness in science and technology, as well as for the national economy. The main goal of the project, which is to measure the Rydberg constant with circular Rydberg atoms, will help resolve the "proton radius puzzle." The puzzle has arisen from inconsistencies in measurements on low-lying electronic and muonic states of hydrogen, which involve the electromagnetic interaction, QED, the proton structure, and QCD. The project will therefore serve as an indirect test for QED and the structure of the proton. With Rydberg atoms, the Rydberg constant will be tested at electron energies in the range of milli-electronvolts. This regime is rather different from the energy regime of previous measurements (eVs to keVs). More generally, compact two-frequency laser sources can bring Rydberg atoms into space, which carries the potential for novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of new phases of matter in very dilute, but strongly interacting, Rydberg-dressed Bose Einstein Condensates.
	Task Progress The research was concentrated on the investigation of ponderomotive optical lattices for Rydberg atoms and on ways how to drive Rydberg-atom transitions by microwave-modulating Rydberg-atom lattices. Several high-precision measurements of Rydberg atoms were conducted, and related theoretical studies were performed. Near the end of the performance period, the team published on tractor atom interferometry, a novel atom-interferometric method. In the following, we provide a brief timeline of the activities under this grant. The publications referenced provide details on the work performed. [Ed. Note: See Cumulative Bibliography for complete citations.] 2015: In ponderomotive spectroscopy, an amplitude-modulated optical standing wave was employed to probe Rydberg-atom transitions, utilizing a ponderomotive rather than a dipole-field interaction. Here, we engaged nonlinearities in the modulation to drive dipole-forbidden transitions up to the fifth order. We reached transition frequencies approaching the sub-THz regime. We also demonstrated magic-wavelength conditions, which resulted in symmetric spectral lines with a Fourier-limited peak at the line center. Applicability to precision measurement were discussed, which include a measurement of the Rydberg constant using circular Rydberg atoms in optical lattices. The work was published in Phys. Rev. Lett. 115, 163003 (2015). 2016 and 2017: We have conducted a study of the systematics that will affect a measurement of the Rydberg constant using circular Rydberg atoms in an intensity-modulated optical lattice. These states have long lifetimes, as well as

Task Progress:

negligible quantum electrodynamics (QED), and no nuclear-overlap corrections. Due to these advantages, the measurement can yield a Rydberg-constant value that is free of interdependencies with QED calculations and the nuclear-charge distribution. In our 2017 study, we considered circular Rydberg atoms that are trapped in an optical lattice and that are driven using our lattice-modulation technique to perform Doppler-free spectroscopy. We have optimized laser wavelengths and beam geometries to realize magic-trapping conditions. The selected transitions have no first-order Zeeman and Stark corrections, leaving only manageable second-order Zeeman and Stark shifts. For rubidium, the projected relative uncertainty of the Rydberg constant in a measurement under Earth's gravity is 10^−11, with the main contribution to the uncertainty coming from the residual lattice shifts. In a future microgravity implementation, this could be reduced into the low 10^-12 range. The study was published in Phys. Rev. A 96, 032513 (2017) and was reported at the CPEM 2018 in Paris (https://).

2018: To better understand the physics of Rydberg atoms at high spectroscopic precision, we have measured the hyperfine structure of rubidium-85 nS1/2 Rydberg states for n = 43, 44, 45, and 46. From the splittings, the hyperfine coupling constant, AHFS, was determined to be 15.372(80) GHz. This result represented an order-of-magnitude improvement from previous measurements. The work was published in Phys. Rev. A 100, 062515 (2019).

2019 and 2020: A leading challenge in performing spectroscopy and quantum information science with circular Rydberg atoms is to prepare such atoms. While we have done this before with low-frequency electromagnetic fields, a method that dovetails much better with Rydberg atoms trapped in an optical lattice is to use the lattice itself to do the circularization. To this end, we have theoretically explored three schemes of initializing circular-state Rydberg atoms via optical couplings provided by the optical ponderomotive effect. In our first method, a radial optical trap consisting of two Laguerre-Gaussian beams of opposite winding numbers transfers orbital angular momentum to the Rydberg atom, providing a first-order coherent coupling between an F-type Rydberg state and a circular state. Next, we considered a one-dimensional ponderomotive optical lattice modulated at RF frequencies, providing quadrupole-like couplings in the hydrogenic manifold for rapid adiabatic passage through a series of intermediate Rydberg states into the circular state. In the third scheme, a two-dimensional ponderomotive optical lattice with a time-orbiting trap center induces an effective circularly polarized RF field of tunable purity for all-optical rapid adiabatic passage into the circular state. The work was published in Phys. Rev. A 101, 013434 (2020).

Over the course of the project, we have delved deeper into developing a thorough understanding of the sub-Doppler characteristics of lattice-modulation spectroscopy of Rydberg atoms. We developed and studied quantum and semiclassical models of Rydberg-atom spectroscopy in amplitude-modulated lattices. Both initial- and target-state Rydberg atoms were assumed to be trapped in the lattice. Unlike in other spectroscopic schemes, the modulation-induced ponderomotive coupling between the Rydberg states is spatially periodic and perfectly phase-locked to the lattice trapping potentials. This leads to a novel type of sub-Doppler mechanism which we have explained in detail. In our exact quantum model, we solved the time-dependent Schrödinger equation in the product space of center-of-mass (COM) momentum states and the internal-state space. We also developed a perturbative model based on the band structure in the lattice and Fermi's golden rule, as well as a semiclassical trajectory model in which the COM was treated classically and the internal-state dynamics quantum-mechanically. In all models we obtained the spectrum of the target Rydberg-state population versus the lattice modulation frequency. Applications in Rydberg-atom spectroscopy were discussed. The work was published in Phys. Rev. A 101, 033414 (2020).

In our third 2020 study, we continued our push towards higher angular momentum states by measuring the quantum defects of G Rydberg states of rubidium. This is important in view of measuring the Rydberg constant because the quantum defects of moderately high angular momentum states, such as G and H, reveal the atoms' core polarizability, which will be needed for Rydberg-constant measurements at the < 10^-11 level using circular atoms. We utilized two-photon, high-precision microwave spectroscopy of nG→ (n + 2)G transitions to precisely measure the high-angular-momentum G-series quantum defect of 85Rb. Samples of cold Rydberg atoms in the nG state were prepared via a three-photon optical excitation combined with controlled electric-field mixing and probed with microwave pulses. We obtained values for the G quantum defects of delta0 = 0.0039990(21) and = −0.0202(21). The work was published in Phys Rev. A 102, 062817 (2020).

2021: In view of earlier works on atom interferometry, and motivated through various NASA Fundamental Physics (FP) conferences, the team became interested in following up on earlier thoughts about novel methods in atom interferometry. We have devised a method called tractor atom interferometry (TAI). TAI employs uninterrupted 3D confinement of interfering atomic wave packets in user-controlled tractor potentials throughout the AI sequence. TAI provides guaranteed closure, suppresses wave-packet dispersion, and is robust against coherence loss from environmental platform dynamics. It allows for arbitrary user-controlled AI paths for maximal sensitivity to certain observables while suppressing contributions from forces that may not be of interest. Designer tractor trajectories can eliminate satellite rotation effects or implement arbitrary hold times. For instance, TAI can selectively maximize sensitivity to acceleration, or rotation, or the gravity-gradient etc. Using Crank-Nicolson simulation of the time-dependent Schrödinger equation, we computed the quantum evolution of scalar and spinor wave functions in several TAI sample scenarios. The work was published in Phys. Rev. A 104, 013307 (2021).

In another piece of 2021 work, we have investigated subtle yet fundamentally interesting aspects of photoionization of Rydberg atoms in optical lattices. We have developed a formalism for photoionization (PI) and potential energy curves (PECs) of Rydberg atoms in ponderomotive optical lattices and applied it to examples covering several regimes of the optical-lattice depth and atomic angular momentum. Characterization of PI in GHz-deep Rydberg-atom lattices is expected to be beneficial for optical control and quantum-state manipulation of Rydberg atoms, while data on PI in shallower lattices are useful in high-precision spectroscopy, such as measurement of the Rydberg constant, and in quantum-computing applications of lattice-confined Rydberg atoms. The work was published in the New Journal of Physics 23, 063074 (2021).

2022: The project finished up with hyperfine measurements of nP1/2 Rydberg states of Rb-85. We have measured the hyperfine structure of nP1/2 Rydberg states for n = 42, 43, 44, and 46 using cold-atom mm-wave spectroscopy. Our result on the nP1/2 hyperfine coupling constant was AHFS = 1.443(31) GHz. This measurement is expected to be useful in studies of long-range Rydberg molecules, Rydberg electrometry, and quantum simulation with dipole-dipole interactions involving nP1/2 atoms. The work was published in Phys. Rev. A 106, 052810 (2022).

2023: A patent was filed on tractor atom interferometry. See https:// Reference: Raithel, Georg A., and Alisher

	Duspayev, “Tractor Atom Interferometry.” U.S. Patent Application No. 17/891,673. Through the lifetime of the grant, three students graduated (Kaitlin Moore, Ryan Cardman, and Andira Ramos). They all now hold positions in research and development (R&D) companies in quantum sensing (Kaitlin Moore, Ryan Cardman) and in the space industry (Andira Ramos).
Bibliography Type:	Description: (Last Updated: 02/24/2025)
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Patents	US20230056032A1. https://patents.google.com/patent/US20230056032A1/en U.S. Patent Application No. 17/891,673 pending as of February 2023. Feb-2023 Raithel G, Duspayev A. "Tractor atom interferometry. "